

CURRICULUM VITAE

Shuay Abdullayev, PhD

Postdoctoral Research Associate

Complex Carbohydrate Research Center, College of Pharmacy and Biomedical Sciences, University of Georgia, Athens, Georgia, United States

Contact and identifiers

Full name (as in publications)	Shuay Abdullayev
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Google Scholar	Shuay Abdullayev
Scientific field	Organic chemistry; carbohydrate chemistry

Profile summary

Shuay Abdullayev is a postdoctoral research associate at the Complex Carbohydrate Research Center of the University of Georgia in Athens, Georgia, in the group of Prof. David Crich. He received a PhD in organic chemistry from the Université du Québec à Montréal in 2021 for a thesis on the synthesis of therapeutic glyconanomaterials, supervised by Prof. René Roy. He previously studied chemistry at Baku State University and at the Université Claude Bernard Lyon 1. His listed papers concern galectin ligands, sialylated oligosaccharides, neoglycolipids, and related carbohydrate chemistry. The CV states an h-index of 4.

Education

Years	Degree	Field / thesis title	Institution, city, country
2016–2021	PhD	Organic chemistry; thesis: New strategy toward the synthesis of therapeutic glyconanomaterials; supervisor: Prof. René Roy	Université du Québec à Montréal (UQAM), Montréal, Canada
2014–2016	MSc	Organic synthesis and chemistry of bioactive molecules	Université Claude Bernard Lyon 1, Lyon, France
2012–2014	BSc (Licence 2 and 3)	Chemistry	Université Claude Bernard Lyon 1, Lyon, France
2007–2011	BSc	Applied chemistry	Baku State University, Baku, Azerbaijan

Academic and professional positions

Years	Position	Department / unit	Institution, city, country
2022–present	Postdoctoral Research Associate	Complex Carbohydrate Research Center	University of Georgia, Athens, Georgia, United States; supervisor: Prof. David Crich
2021–2022	Research Scientist		Zamboni Chem Solution Inc., Montréal, Canada; director: Helmi Zaghdane

Years	Position	Department / unit	Institution, city, country
2017–2020	Teaching Assistant	Bio-organic chemistry; organic chemistry laboratory	Université du Québec à Montréal, Montréal, Canada
2017	ChemNet Bootcamp trainee (2 weeks)		Structural Genomics Consortium, Toronto, Canada
2016	Research Assistant (6 months)	Crich Group, Department of Chemistry	Wayne State University, Detroit, United States; supervisor: Prof. David Crich. Project: mechanism- based exploration of methods for the diastereoselective synthesis of glycosidic bonds, with emphasis on sialic acid glycosides and α -C- sialidation
2015	Intern (2 months)	Institut de Chimie et Biochimie Moléculaire et Supramoléculaire (ICBMS)	Villeurbanne, France; supervisor: Dr. Sébastien Vidal. Project: tetraphenylethylene-based glycoclusters with aggregation- induced emission as ligands of bacterial lectins
2010	Intern (2 months)	Babanly Group, Department of Chemistry	Baku State University, Baku, Azerbaijan; supervisor: Prof. Mahammad Babanly. Project: synthesis of binary and ternary semiconductor compounds and their physicochemical analysis

Research metrics

Indicator	Value	Source	Date checked
h-index	4	Google Scholar	

Publications

Peer-reviewed journal articles

- Massaro, M.; Cagnoni, A. J.; Medrano, F. J.; Perez-Saez, J. M.; **Abdullayev, S.**; Belkhadem, K.; Marino, K. V.; Romero, A.; Roy, R.; Rabinovich, G. A. Selective modifications of lactose and N-acetyllactosamine with sulfate and aromatic bulky groups unveil unique structural insights in galectin-1-ligand recognition. *Bioorg. Med. Chem.*, 2023, 94, 117480.
- Abdullayev, S.**; Kadav, P.; Bandyopadhyay, P.; Medrano, F. J.; Rabinovich, G. A.; Dam, T. K.; Romero, A.; Roy, R. Selectively modified lactose and N-acetyllactosamine analogs at three key positions to afford effective galectin-3 ligands. *Int. J. Mol. Sci.*, 2023, 24, 3718.
- Mousavifar, L.; Lewicky, J. D.; Taponard, A.; Bagul, R.; Rivat, M.; **Abdullayev, S.**; Martel, A. L.; Fraleigh, N. L.; Nakamura, A.; Veyrier, F. J.; Le, H.-T.; Roy, R. Synthesis and evaluation of novel mannosylated neoglycolipids for liposomal delivery system applications.
- Abdullayev, S.**; Roy, R. Practical non-enzymatic synthesis of propargyl sialyl- α -(2-3')-lactosamine trisaccharide using minimal protecting groups manipulation. *Carbohydr. Res.*, 2022, 514, 108543.
- Mélin, L.; **Abdullayev, S.**; Fnaiche, A.; Vu, V.; Gonzalez-Suarez, N.; Zeng, H.; Szewczyk, M.; Li, F.; Senisterra, G.; Allali-Hassani, A.; Chau, I.; Dong, A.; Woo, S.; Annabi, B.; Halabelian, L.; LaPlante, S.; Vedadi, M.; Barste-Lovejoy, D.; Santhakumar, V.; Gagnon, A. Development of LM98, a small-molecule TEAD inhibitor derived from flufenamic acid. *ChemMedChem*, 2022, 16, 2982.
- Mousavifar, L.; **Abdullayev, S.**; Roy, R. Recent development in the design of neoglycoliposomes bearing arborescent architectures. *Molecules*, 2021, 26, 4281.

7. Ogasahara, R.; **Abdullayev, S.**; Sarpe, V. A.; Mandhapat, A. R.; Crich, D. Influence of protecting groups on O- and C-glycosylation with neuraminy and ulosonyl dibutylphosphates. *Carbohydr. Res.*, 2020, 496, 108100. Equal contribution.

8. Donnier-Maréchal, M.; **Abdullayev, S.**; Bauduin, M.; Pascal, Y.; Fu, M-Q.; He, X-P.; Gillon, E.; Imbert, A.; Kipnis, E.; Dessein, R.; Vidal, S. Tetraphenylethylene-based glycoclusters with aggregation induced emission (AIE) properties as high affinity ligands of bacterial lectins. *Org. Biomol. Chem.*, 2018, 16, 8804.

Awards, honors and fellowships

Year	Award / honor	Awarding body
2019	Concours de bourses d'études supérieures Pharmaqam (UQAM scholarship)	UQAM, Canada
2015	Ministry of Education scholarship	Ministry of Education of Azerbaijan
2014	Bourse du Gouvernement Français (BGF – Sécurité Sociale)	French Government
2013	Bourse du Gouvernement Français (BGF)	French Government
2012	Ministry of Education scholarship	Ministry of Education of Azerbaijan

Teaching

Years	Course	Level	Institution
2017–2020	Bio-organic chemistry; organic chemistry laboratory	Teaching assistant	Université du Québec à Montréal

Invited talks and conference participation

Year	Event, place	Type	Title of talk
2018	Symposium du Réseau de l'Université du Québec sur la Recherche Biomédicale et Biopharmaceutique, Sainte-Adèle, Canada	Poster	
2016	27th Québec/Ontario Mini-symposium for Synthetic and Bioorganic Chemistry (QOMSBOD), University of Waterloo, Waterloo, Canada	Oral	

Languages

Language	Level
Azerbaijani	Native
Turkish	Fluent
French	Fluent
English	Fluent
Russian	Average
Spanish	Beginner